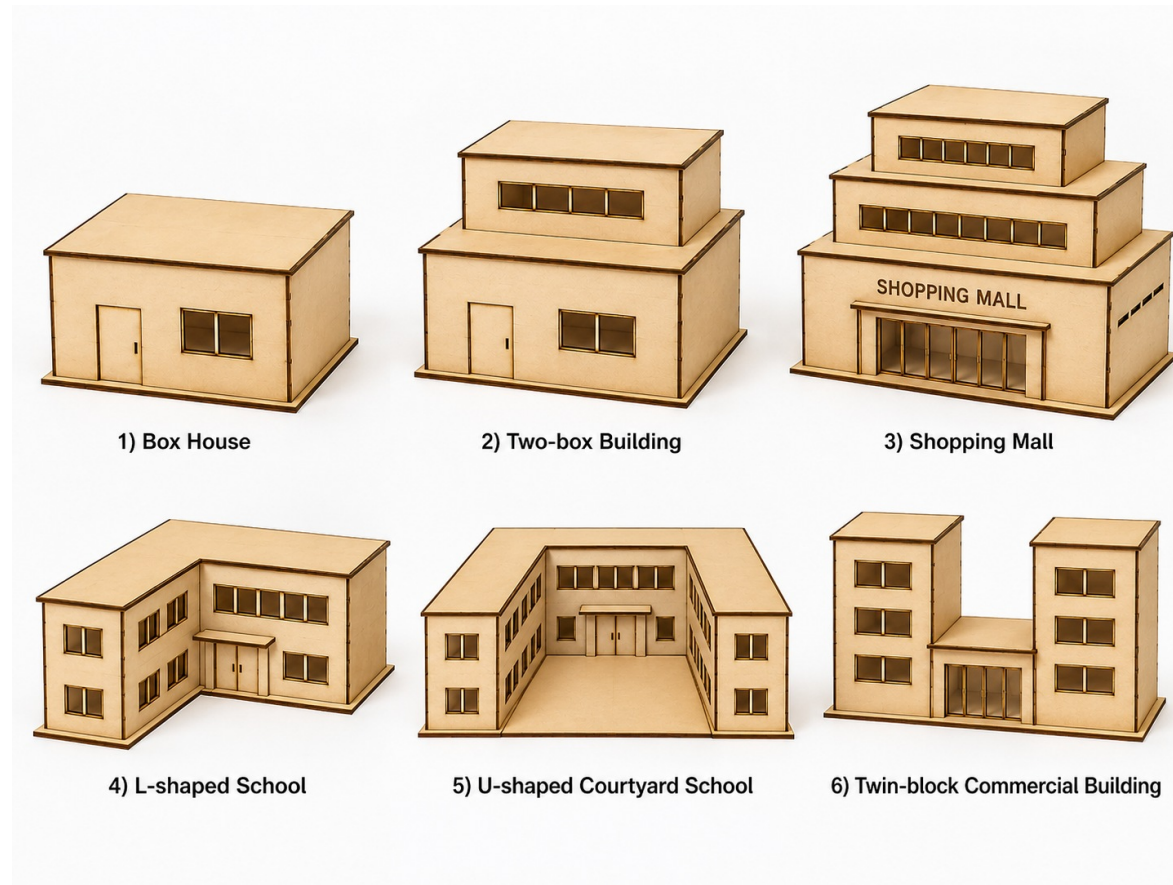


Build Hong Kong buildings with wood

CorelDRAW 2019 (Windows) | Form 2 | 2 mm wood



Draw flat parts. Assemble a 3D building.

Start with W1, then choose a building. Each step: find the tool, do the action, check the result.

Class: _____ Group: _____ Name: _____

Choose a model: start with W1

Model menu

Click a picture to jump to its dimensions. New to CorelDRAW? Complete W1 first.



W1 Single-storey house

80 × 60 × 50 mm | 6 panels



W2 Two-level stepped building

80 × 60 × 80 mm | 12 panels



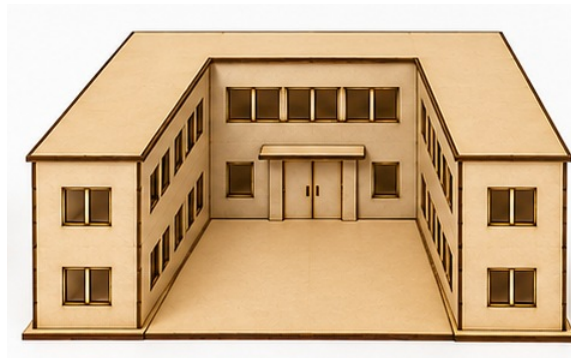
W3 Three-level mall

100 × 70 × 95 mm | 18 panels



W4 L-shaped school

100 × 100 × 40 mm | 12 panels



W5 U-shaped school

120 × 100 × 40 mm | 18 panels



W6 Twin-block commercial building

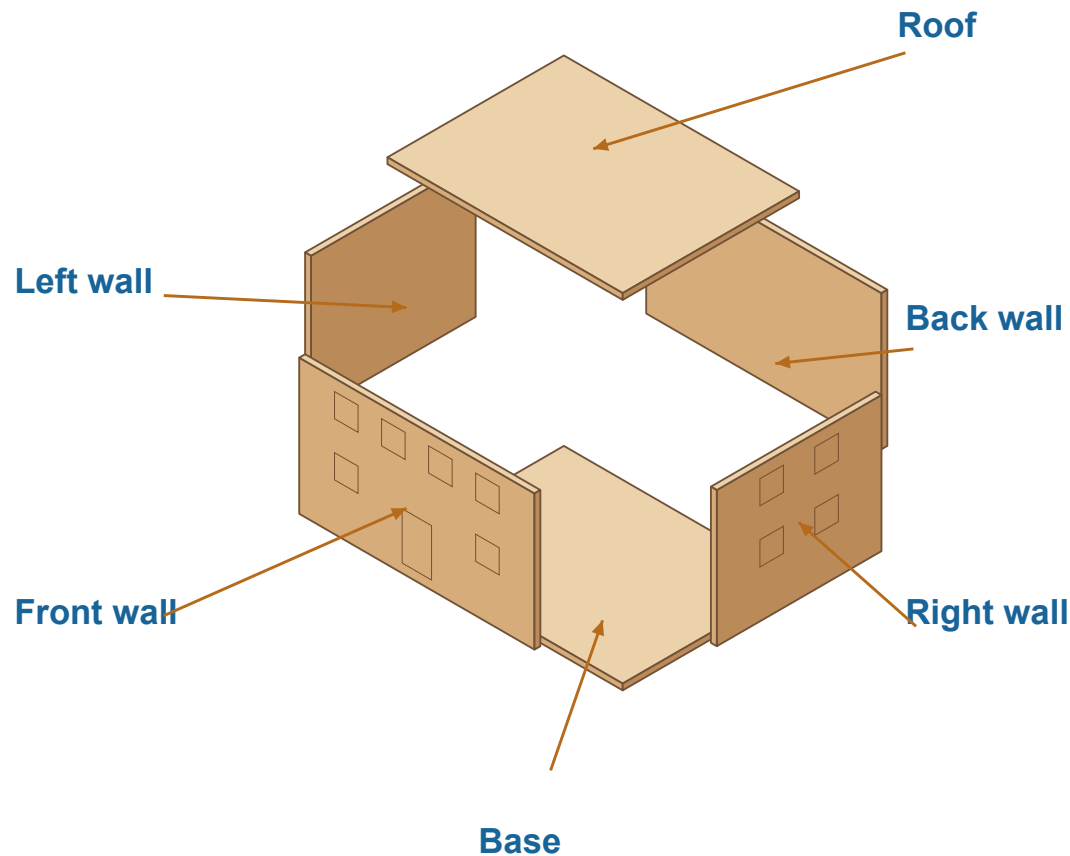
160 × 60 × 90 mm | 18 panels

Pictures show the style. Use this guide for sizes: straight edges, flat roofs and engraved doors and windows.

How six panels make a house

Panel joints

W1 finished outer size: width 80 × depth 60 × height 50 mm.



One roof and one base

80 × 60 mm. These cover the top and bottom of all four walls.

One front and one back wall

80 × 46 mm.

One left and one right wall

56 × 46 mm.

Exploded view: gaps separate the parts for viewing only.

Side walls fit BETWEEN the front and back walls. The listed sizes already allow for wood thickness. Do not subtract it again.

01 | New document: use millimetres

Basic skills

The page is your layout area. It is not the building size or the laser machine bed.



Start in the File menu. The table shows the settings.

Document settings	Enter
Drawing units	Millimetres (mm)
Page width	300 mm
Page height	200 mm

Press Ctrl + N

Or choose File > New.

Set the units and page

Choose mm. Set width to 300 and height to 200. Click OK.

Check the result

You should see a blank landscape page. Now draw the first wall.

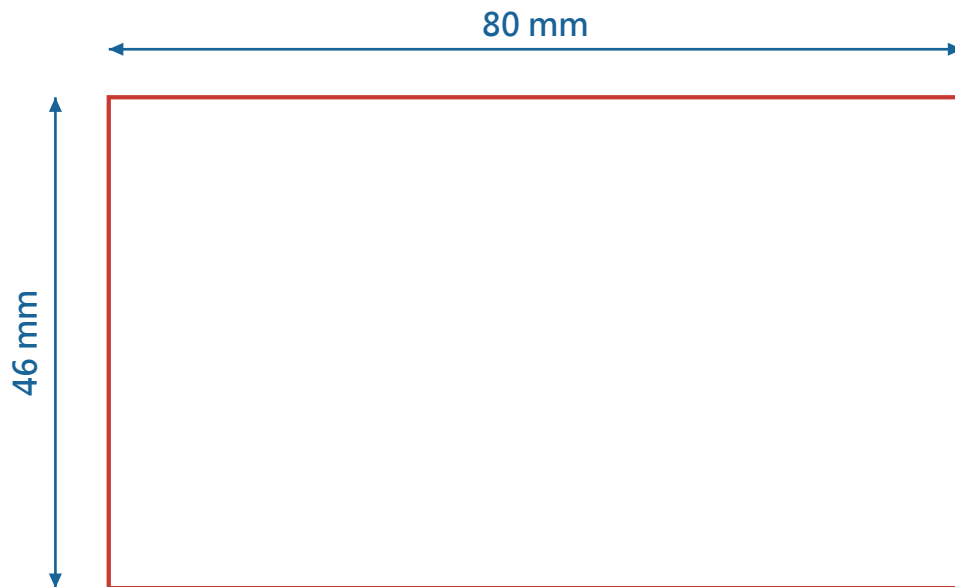
02 | First wall: 80 × 46 mm

Basic skills

Draw a rough rectangle, then enter its size. Your mouse does not need to be exact.

Property bar diagram (not a live screenshot)

Object size	W: 80 mm	Unlock aspect ratio
	H: 46 mm	



Front / back wall

Press F6; drag with the left button

On the white page, hold the left mouse button. Drag from one corner to another, then release.

Set the object size

Select the rectangle. Unlock the aspect ratio on the property bar. Enter width 80 and height 46; press Enter after each.

Check both numbers

Select it again: it must be 80 × 46 mm. If one value changes the other, unlock the ratio and re-enter both.

03 | No fill: keep the outline

Basic skills

Classroom colours: red outlines = cutting; blue lines = surface engraving.



Official palette detail: find these swatches



LEFT-click the No Color swatch.
Keep the inside transparent.



RIGHT-click the red swatch.
The outline marks a cut.



Left

Select the wall

Choose the black-arrow Pick tool. Left-click the rectangle outline.



Left

Remove the fill

On the right palette, LEFT-click the No Color (×) swatch. Use the mouse, not the X key.



Right

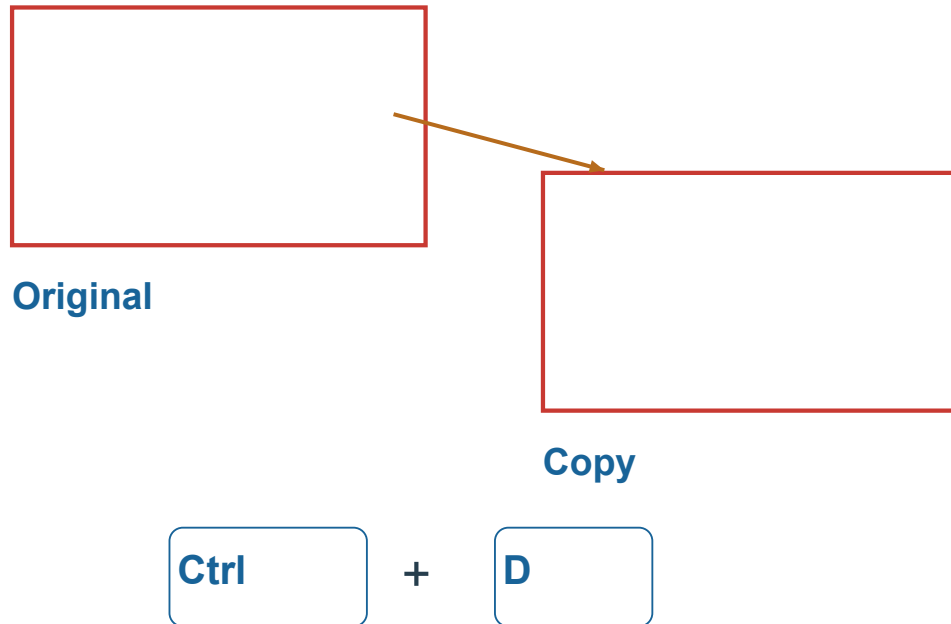
Set the outline

RIGHT-click red. If the whole shape turns red, press Ctrl+Z and try the right button. Use the line width your teacher specifies.

04 | Duplicate: make all six panels

Basic skills

Move each copy apart so identical cutting outlines do not overlap.



A copy may already be offset. Use the Pick tool and drag its outline apart with the left button.

Select the front wall; Ctrl + D

This makes the back wall: another 80 × 46 mm rectangle.

Duplicate and resize

Make two more copies. Set each to 56 × 46 mm for the left and right walls.

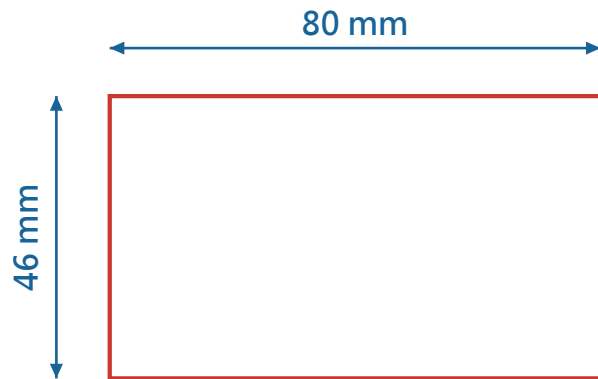
Draw the roof and base

Press F6. Draw 80 × 60 mm with no fill and a red outline. Duplicate it once. You now have six panels.

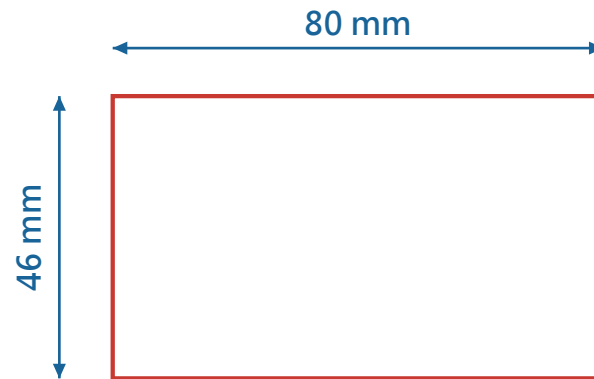
05 | Pause: do you have six panels?

Make W1

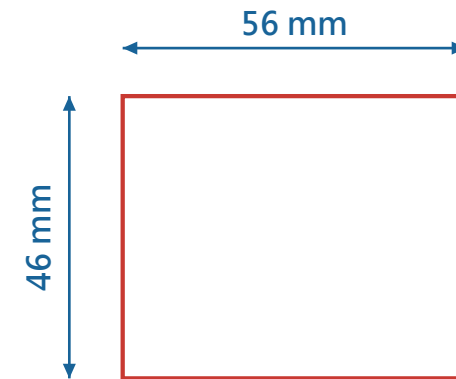
Dimensions for reference. This page is not a ready-to-cut layout file.



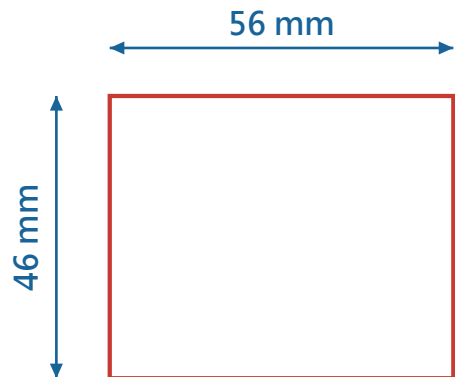
Front wall



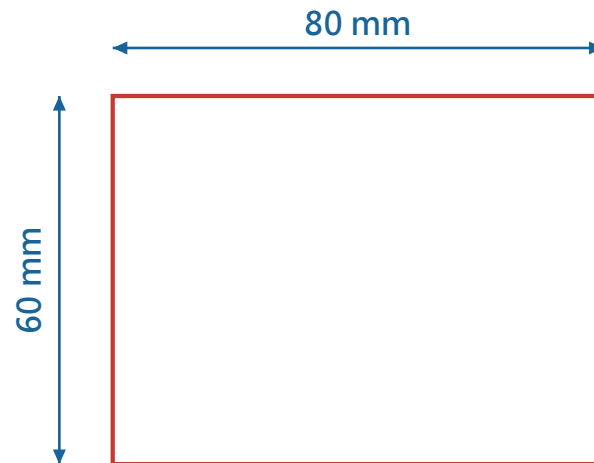
Back wall



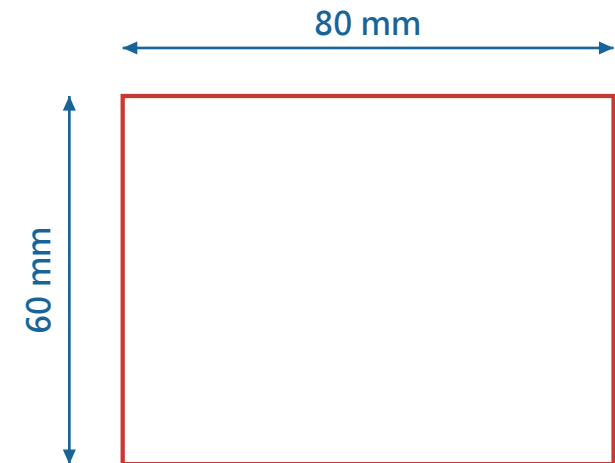
Left wall



Right wall



Roof

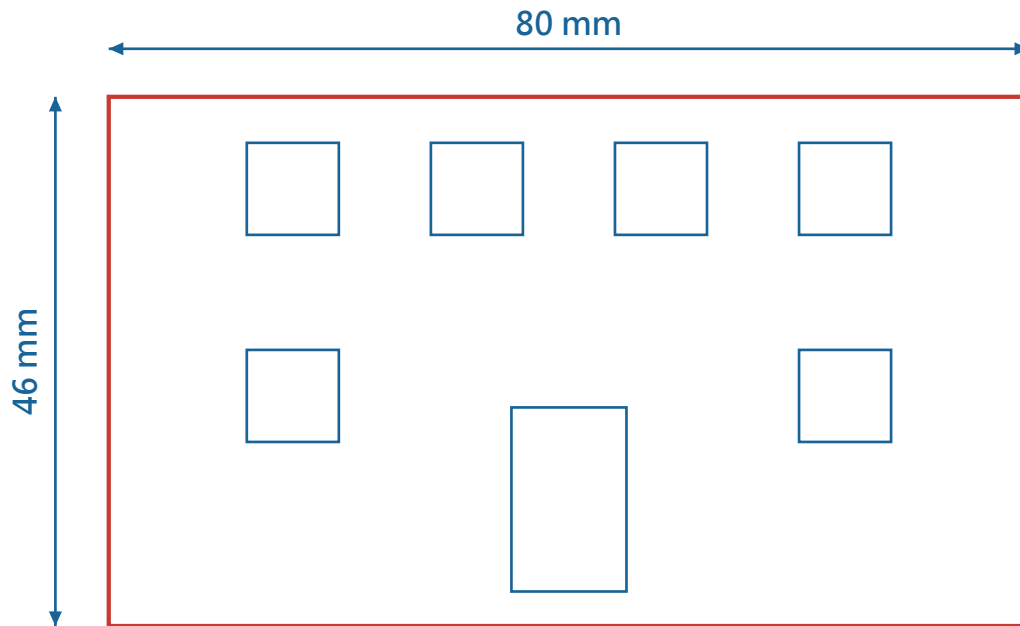


Base

06 | Doors and windows: draw, then copy

Basic skills

Blue rectangles are surface engraving lines. They do not cut the windows out.



W1 front wall: engrave doors and windows

Window: 8 × 8 mm

Door: 10 × 16 mm

Keep every line inside the wall.

Window: press F6

Draw a rectangle, 8 × 8 mm. Left-click No Color; right-click blue.

Use the Pick tool to move it

Place it near the top of the front wall. Press Ctrl+D and move each copy apart. Stay at least 3 mm from outer edges.

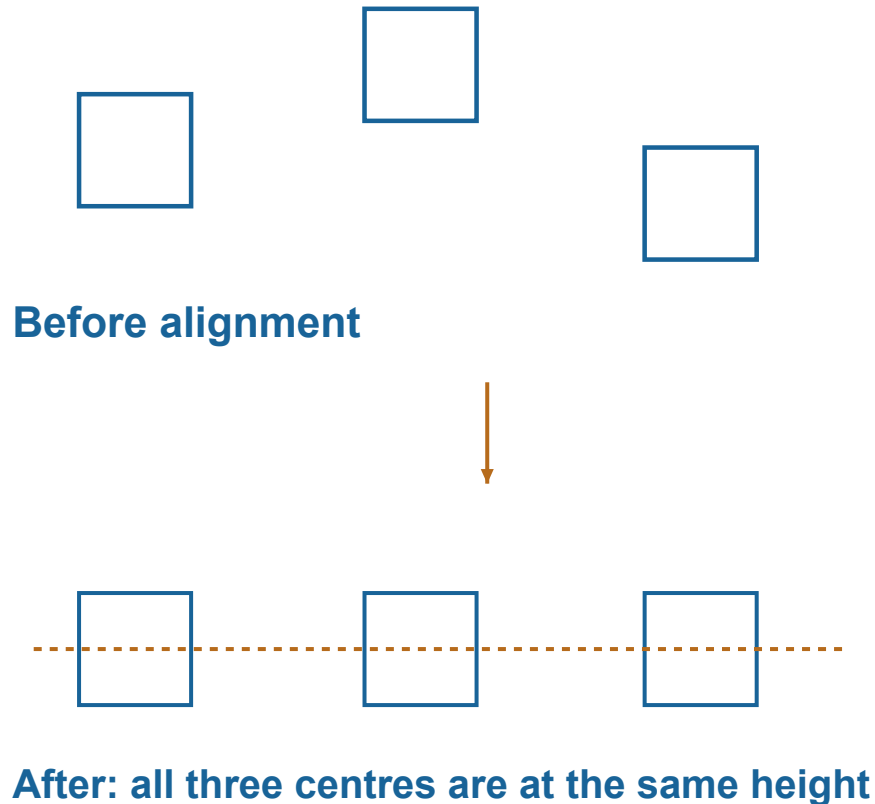
Door: 10 × 16 mm

Use a blue outline near the bottom of the front wall. Leave at least 3 mm below it. You may design your own door and window layout.

07 | Align windows at the same height

Basic skills

Finish entering numbers before selecting objects and using alignment keys.



Select only three windows

Click a window to move. Hold Shift and select another. Select the correctly placed reference window last.

Press E: align centres vertically

Their centres move to the same height. Do not include the wall in the selection: it could move too.

Adjust the gaps

Click blank space, then select one window with the Pick tool. Hold Ctrl while dragging sideways to keep its height. Undo with Ctrl+Z.

08 | Add a sign (optional)

Finish all six panels first. The templates do not contain text.



MY SCHOOL

F8



Ctrl + Q

**Keep an editable CDR copy first.
Save another copy and convert its text to
curves.**

Press F8; left-click once

Type a short sign, such as MY SCHOOL.
Switch to the Pick tool to move it.

Keep the sign inside the wall

Use a blue outline, or the engraving method
your teacher specifies. Keep clear of doors,
windows and outer edges.

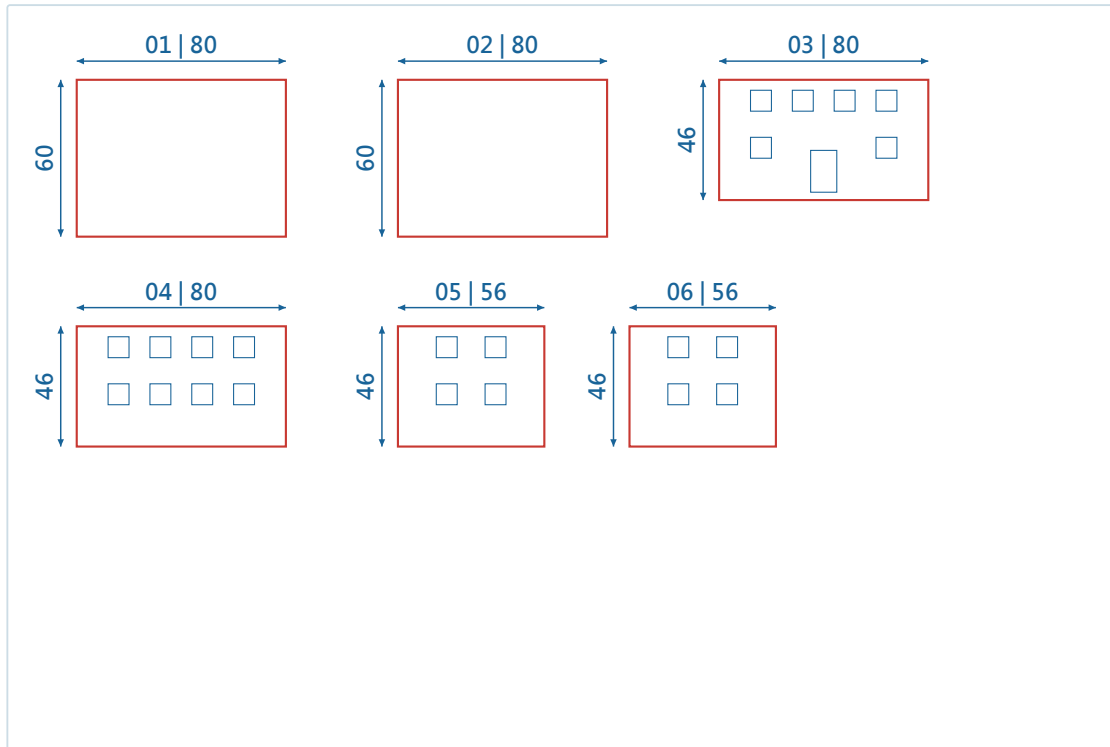
Convert text in the submission copy

Save an editable copy, then a submission
copy. Select the text and press Ctrl+Q. You
cannot directly edit the wording after
conversion.

09 | Arrange and save your CDR

Basic skills

One complete model = quantity 1. Do not use the number of panels as the quantity.



W1 layout | Above: number / width; left: height

Units: mm. SVG page: 300 × 200 mm.

Dimension labels are only in this guide, not the cutting file.

Leave space between outlines

Keep all panels on the page, at least 6 mm apart as in the template. Select or group each panel with its door and windows before moving.

Press Ctrl + S to save

Choose CorelDRAW (CDR).
Example: 2B_GROUP 2_W1_1.cdr

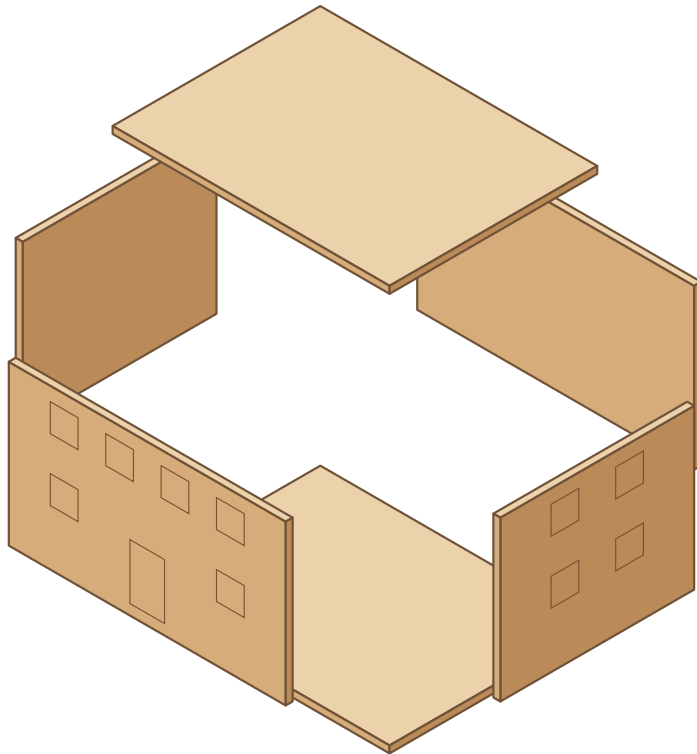
For a two-sheet template

Import each SVG into a separate new document. Save with _S01 or _S02 at the end.
Example: 2B_GROUP 2_W3_1_S01.cdr
Include a full-view screenshot of each file.

After cutting: test-fit before gluing

Assembly

Your teacher arranges cutting. Use the glue and assembly method your teacher recommends.



1 | Identify all six parts

Find front, back, left, right, roof and base. The front wall has the door. Do not glue yet.

2 | Put four walls on the base

Fit the side walls between the front and back walls. Hold them with tape. Check that all corners are square.

3 | Test the roof, then glue

The roof should cover all four walls. If it fits, glue the joints one at a time. Let the glue dry before stacking boxes.

Roof does not fit? Check that side walls sit between front and back walls, and that you have not used the roof as a wall.

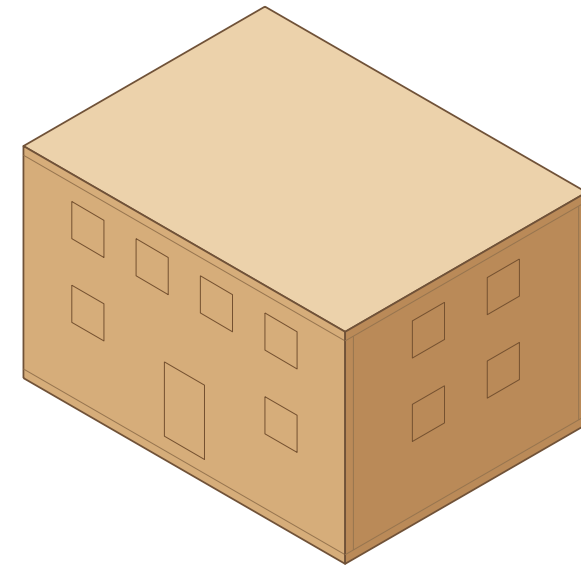
Follow your teacher when using hot glue or tools. Do not touch hot glue or material while the laser is working.

W1 | Single-storey house

Finished outer size: W 80 × D 60 × H 50 mm; 6 panels.



Style reference: adjust the number of windows to suit your time



Assembly diagram: use the box sizes below

Box	Use	Outer W × D × H (mm)
A	Main box	80 × 60 × 50

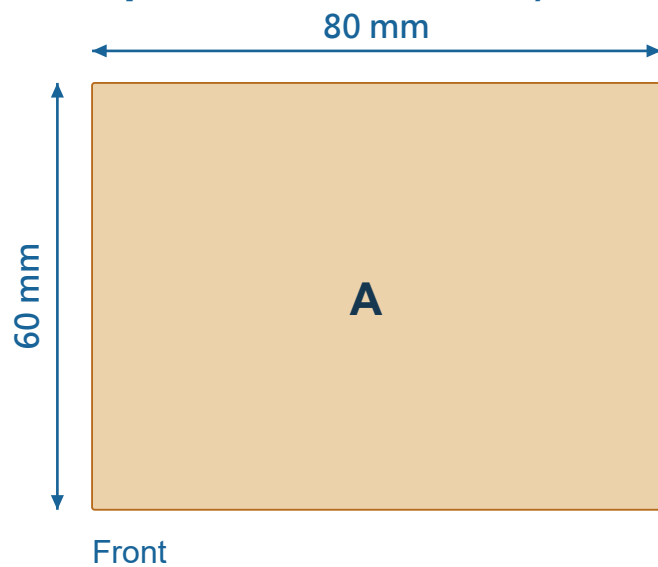
W1 | Draw the panels

Panel sizes

All sizes in mm, adjusted for 2 mm wood. Make TWO of each listed size. Do not subtract thickness again.

Box	Front / back × 2	Left / right × 2	Roof / base × 2
A Main box	80 × 46	56 × 46	80 × 60

Box positions (top view)



One box

Start with the base, add walls, then the roof. Front and back walls cover the ends of the side walls.

Finished height: 50 mm

46 mm walls, plus a 2 mm roof and a 2 mm base.

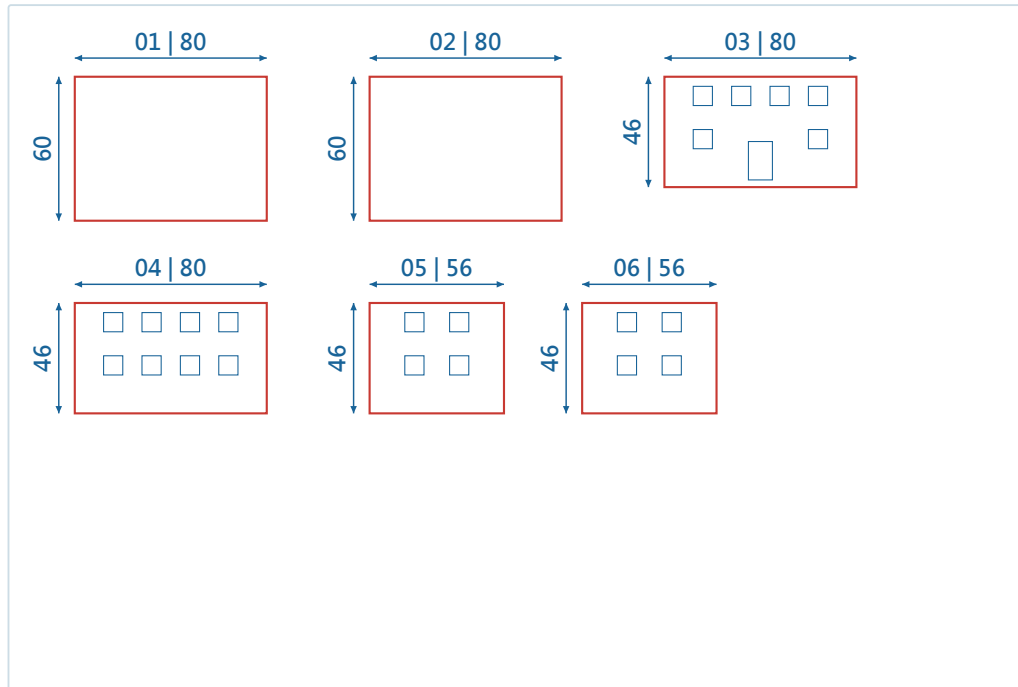
Blue lines engrave doors and windows. Leave hidden joining faces blank. Diagrams are scaled to fit; use the stated dimensions.

W1 | Template layout and part key

Template key

Folder: templates/W1 | 1 SVG sheet(s) | Each page: 300 × 200 mm.

Sheet 1



Above: part number / width; left: height. Units: mm. Spacing adjusted for clarity; cutting SVGs have no labels.

01 A Base 80 × 60 mm

02 A Roof 80 × 60 mm

03 A Front 80 × 46 mm

04 A Back 80 × 46 mm

05 A Left 56 × 46 mm

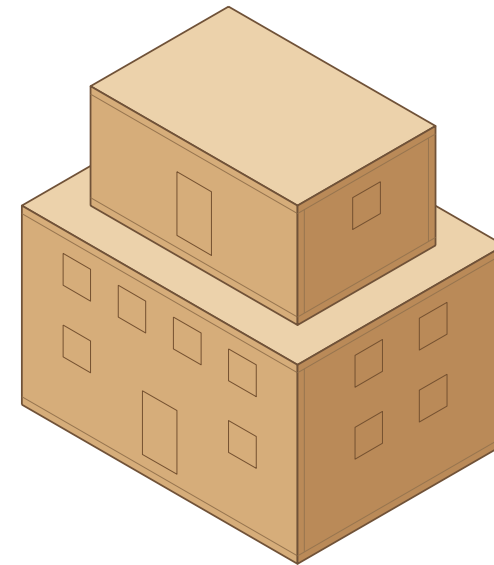
06 A Right 56 × 46 mm

W2 | Two-level stepped building

Finished outer size: W 80 × D 60 × H 80 mm; 12 panels.



Style reference: adjust the number of windows to suit your time



Assembly diagram: use the box sizes below

Box	Use	Outer W × D × H (mm)
A	Lower box	80 × 60 × 50
B	Upper box	60 × 40 × 30

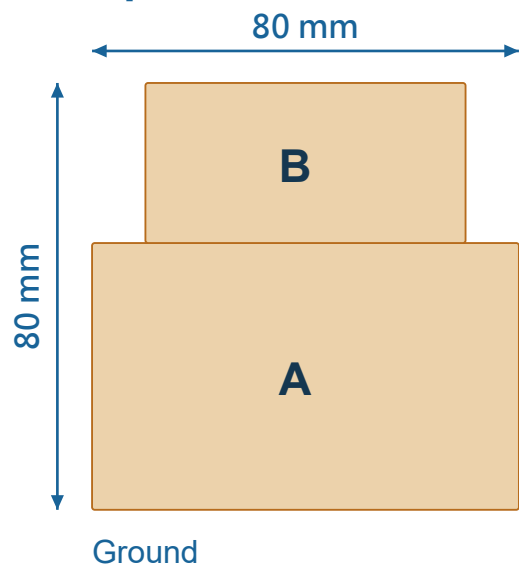
W2 | Draw the panels

Panel sizes

All sizes in mm, adjusted for 2 mm wood. Make TWO of each listed size. Do not subtract thickness again.

Box	Front / back × 2	Left / right × 2	Roof / base × 2
A Lower box	80 × 46	56 × 46	80 × 60
B Upper box	60 × 26	36 × 26	60 × 40

Box positions (front view)



A below, B above

Build two complete six-panel boxes. Centre B on the roof of A.

Inset 10 mm on all sides

B is 20 mm smaller in width and depth than A. Total height: $50 + 30 = 80$ mm.

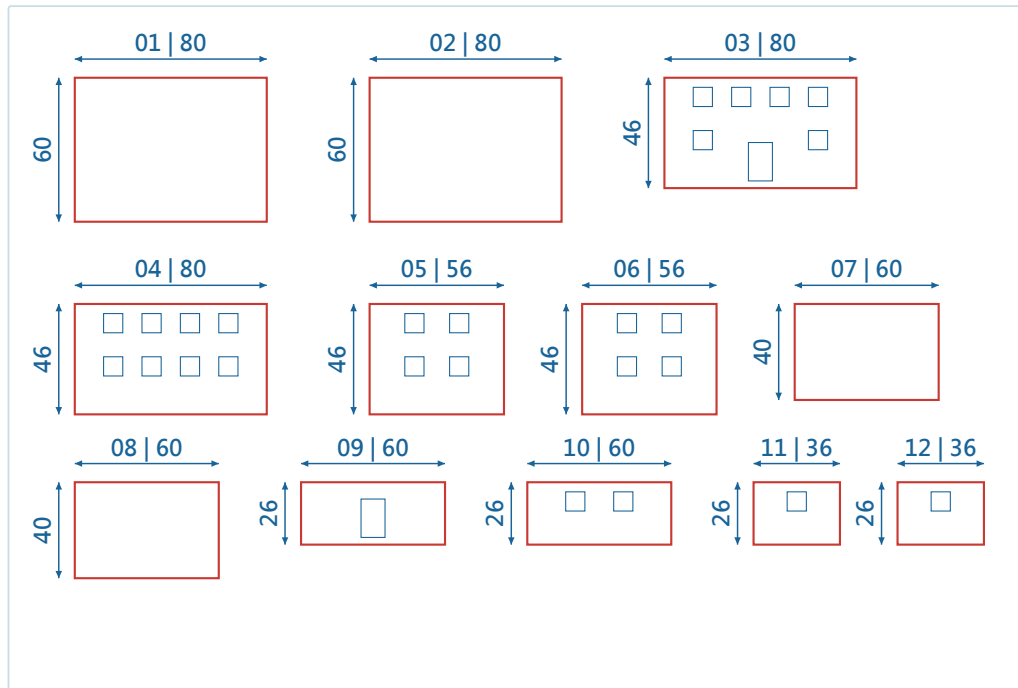
Blue lines engrave doors and windows. Leave hidden joining faces blank. Diagrams are scaled to fit; use the stated dimensions.

W2 | Template layout and part key

Template key

Folder: templates/W2 | 1 SVG sheet(s) | Each page: 300 × 200 mm.

Sheet 1



Above: part number / width; left: height. Units: mm. Spacing adjusted for clarity; cutting SVGs have no labels.

01 A Base 80 × 60 mm

02 A Roof 80 × 60 mm

03 A Front 80 × 46 mm

04 A Back 80 × 46 mm

05 A Left 56 × 46 mm

06 A Right 56 × 46 mm

07 B Base 60 × 40 mm

08 B Roof 60 × 40 mm

09 B Front 60 × 26 mm

10 B Back 60 × 26 mm

11 B Left 36 × 26 mm

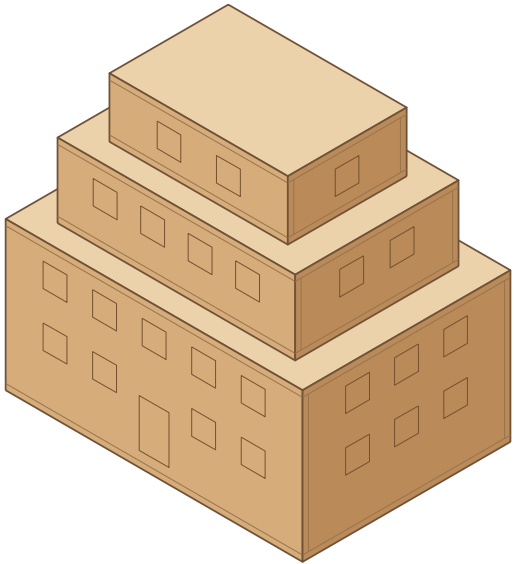
12 B Right 36 × 26 mm

W3 | Three-level mall

Finished outer size: W 100 × D 70 × H 95 mm; 18 panels.



Style reference: adjust the number of windows to suit your time



Assembly diagram: use the box sizes below

Box	Use	Outer W × D × H (mm)
A	Lower box	100 × 70 × 50
B	Middle box	80 × 55 × 25
C	Upper box	60 × 40 × 20

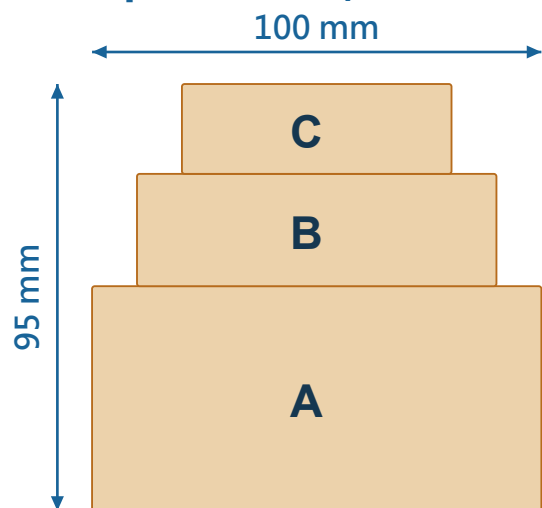
W3 | Draw the panels

Panel sizes

All sizes in mm, adjusted for 2 mm wood. Make TWO of each listed size. Do not subtract thickness again.

Box	Front / back × 2	Left / right × 2	Roof / base × 2
A Lower box	100 × 46	66 × 46	100 × 70
B Middle box	80 × 21	51 × 21	80 × 55
C Upper box	60 × 16	36 × 16	60 × 40

Box positions (front view)



Ground

Stack largest to smallest

Build each box first. Stack A (bottom), B (middle), then C (top).

Centre each layer

Each layer is inset 10 mm at each side and 7.5 mm at front and back. Total height: 95 mm.

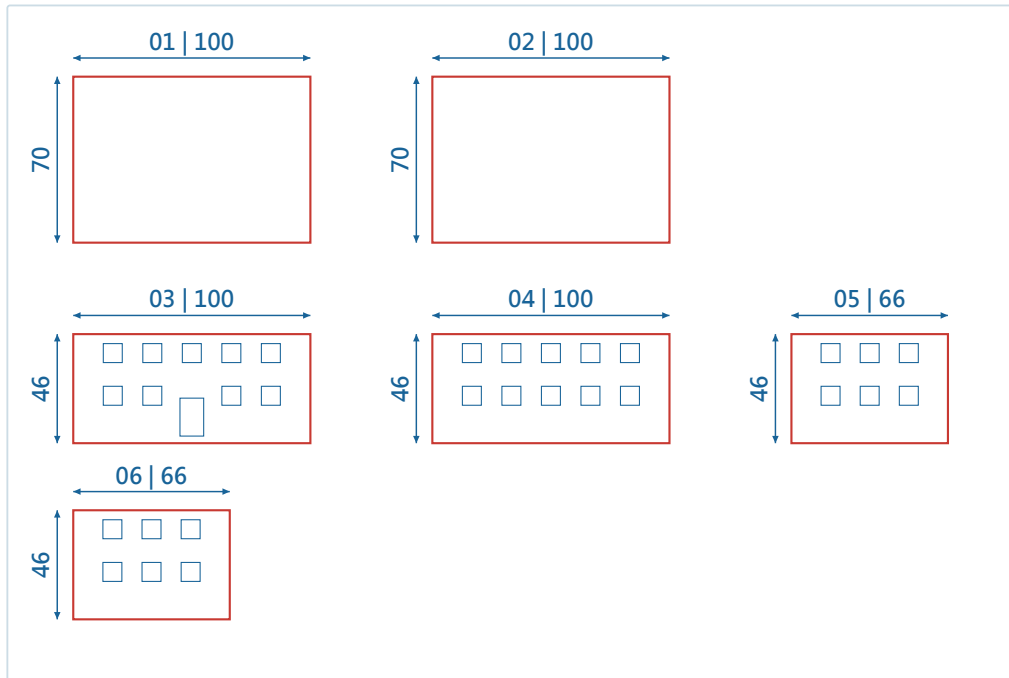
Blue lines engrave doors and windows. Leave hidden joining faces blank. Diagrams are scaled to fit; use the stated dimensions.

W3 | Template layout and part key

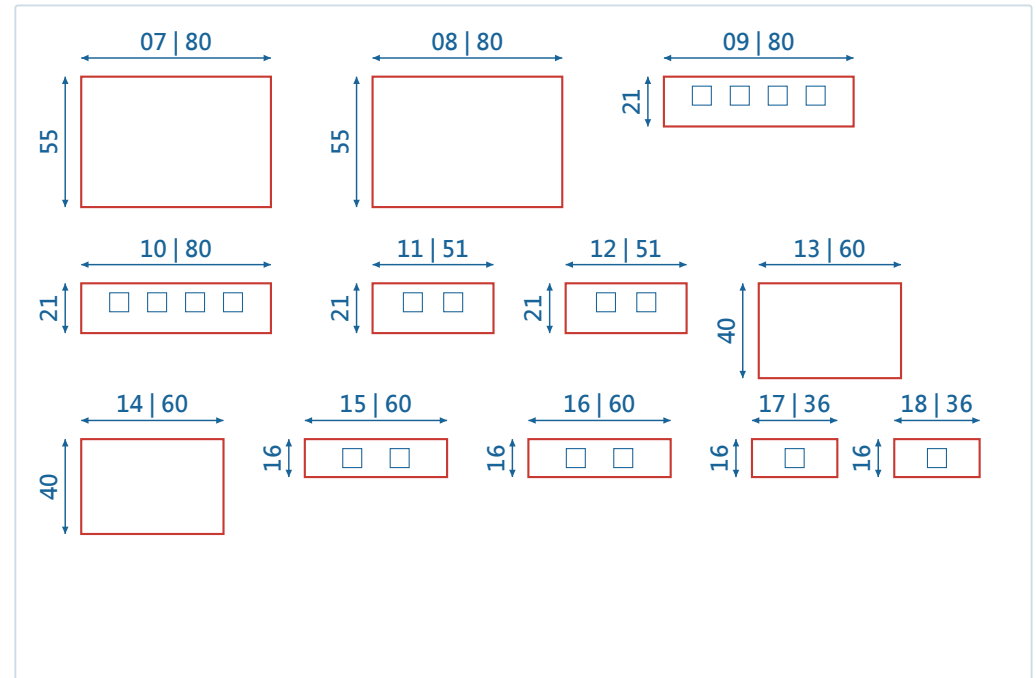
Template key

Folder: templates/W3 | 2 SVG sheet(s) | Each page: 300 × 200 mm.

Sheet 1



Sheet 2



Above: part number / width; left: height. Units: mm. Spacing adjusted for clarity; cutting SVGs have no labels.

01 A Base 100 × 70 mm

02 A Roof 100 × 70 mm

03 A Front 100 × 46 mm

04 A Back 100 × 46 mm

05 A Left 66 × 46 mm

06 A Right 66 × 46 mm

07 B Base 80 × 55 mm

08 B Roof 80 × 55 mm

09 B Front 80 × 21 mm

10 B Back 80 × 21 mm

11 B Left 51 × 21 mm

12 B Right 51 × 21 mm

13 C Base 60 × 40 mm

14 C Roof 60 × 40 mm

15 C Front 60 × 16 mm

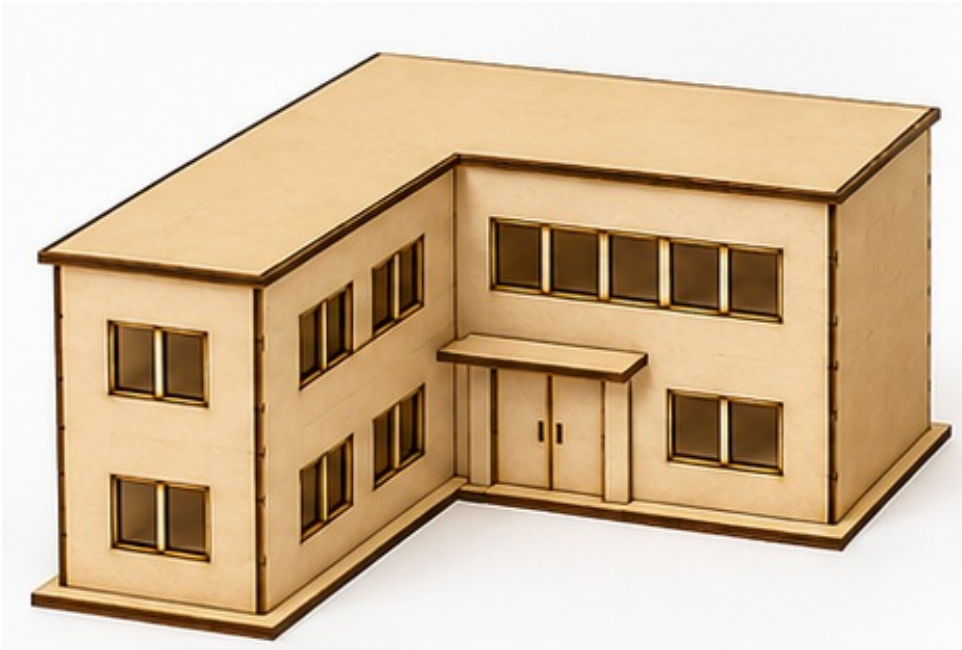
16 C Back 60 × 16 mm

17 C Left 36 × 16 mm

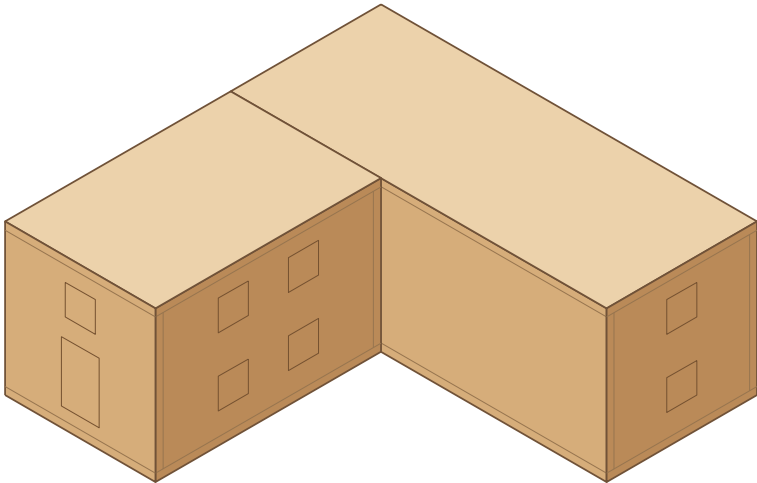
18 C Right 36 × 16 mm

W4 | L-shaped school

Finished outer size: W 100 × D 100 × H 40 mm; 12 panels.



Style reference: adjust the number of windows to suit your time



Assembly diagram: use the box sizes below

Box	Use	Outer W × D × H (mm)
A	Rear block	100 × 40 × 40
B	Left wing	40 × 60 × 40

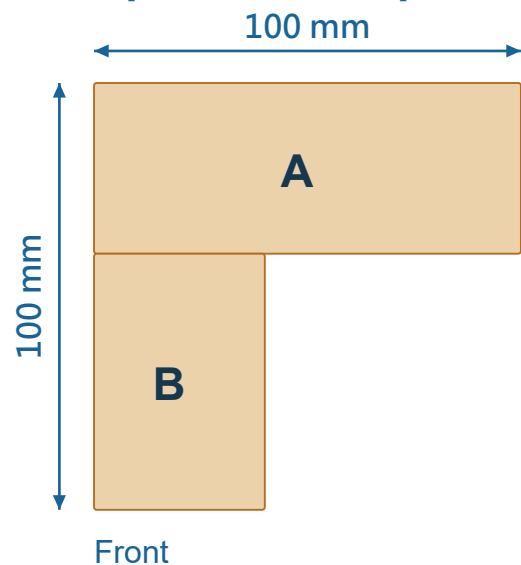
W4 | Draw the panels

Panel sizes

All sizes in mm, adjusted for 2 mm wood. Make TWO of each listed size. Do not subtract thickness again.

Box	Front / back × 2	Left / right × 2	Roof / base × 2
A Rear block	100 × 36	36 × 36	100 × 40
B Left wing	40 × 36	56 × 36	40 × 60

Box positions (top view)



A at the back; B at front left

Place the back of B against the front of A. Align their left edges. Do not overlap the boxes.

Keep all walls of both boxes

The layout makes an L. Leave the front-right space for a playground. Total depth: 40 + 60 = 100 mm.

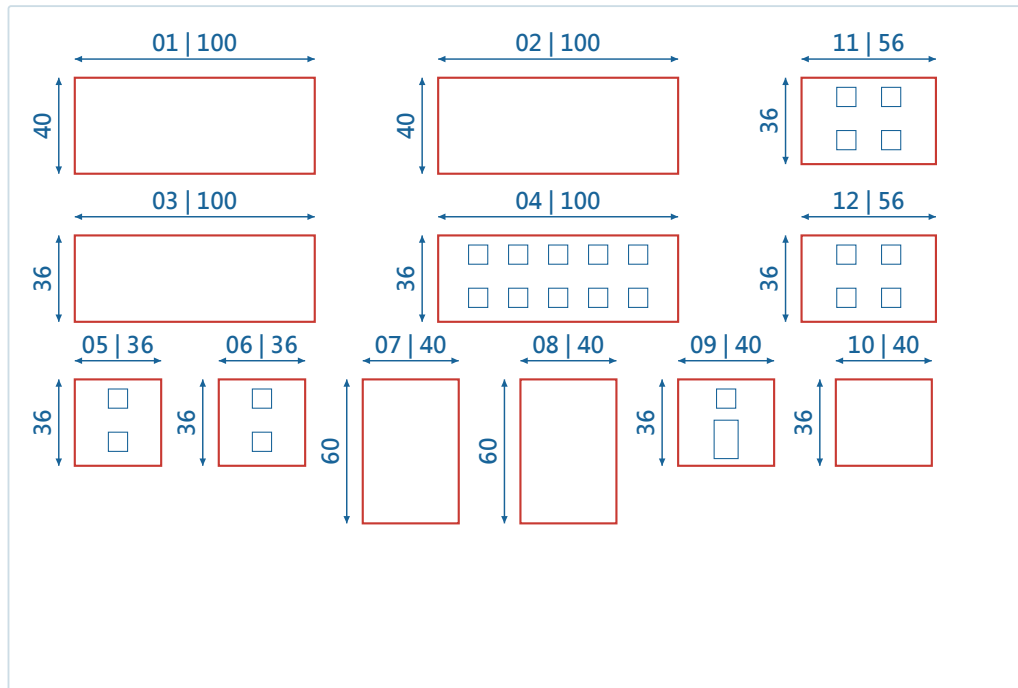
Blue lines engrave doors and windows. Leave hidden joining faces blank. Diagrams are scaled to fit; use the stated dimensions.

W4 | Template layout and part key

Template key

Folder: templates/W4 | 1 SVG sheet(s) | Each page: 300 × 200 mm.

Sheet 1



Above: part number / width; left: height. Units: mm. Spacing adjusted for clarity; cutting SVGs have no labels.

01 A Base 100 × 40 mm

02 A Roof 100 × 40 mm

03 A Front 100 × 36 mm

04 A Back 100 × 36 mm

05 A Left 36 × 36 mm

06 A Right 36 × 36 mm

07 B Base 40 × 60 mm

08 B Roof 40 × 60 mm

09 B Front 40 × 36 mm

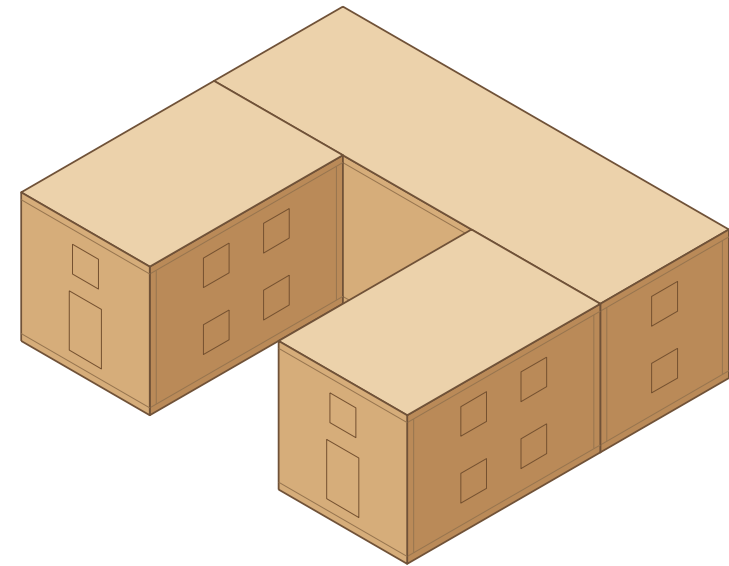
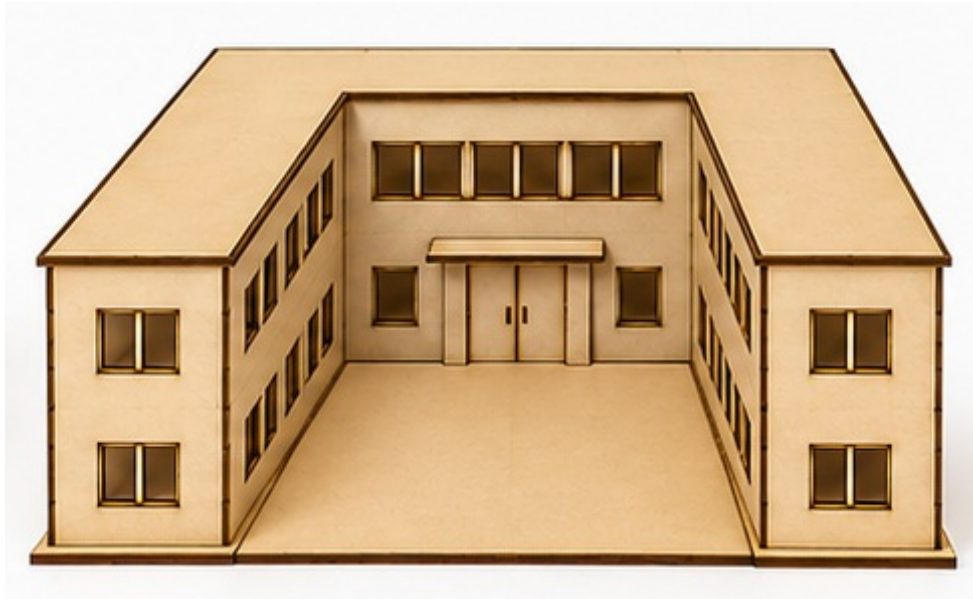
10 B Back 40 × 36 mm

11 B Left 56 × 36 mm

12 B Right 56 × 36 mm

W5 | U-shaped school

Finished outer size: W 120 × D 100 × H 40 mm; 18 panels.



Style reference: adjust the number of windows to suit your time

Assembly diagram: use the box sizes below

Box	Use	Outer W × D × H (mm)
A	Rear block	120 × 40 × 40
B	Left wing	40 × 60 × 40
C	Right wing	40 × 60 × 40

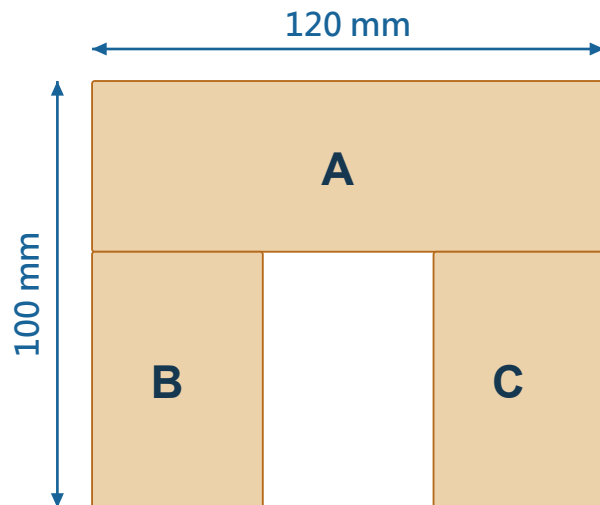
W5 | Draw the panels

Panel sizes

All sizes in mm, adjusted for 2 mm wood. Make TWO of each listed size. Do not subtract thickness again.

Box	Front / back × 2	Left / right × 2	Roof / base × 2
A Rear block	120 × 36	36 × 36	120 × 40
B Left wing	40 × 36	56 × 36	40 × 60
C Right wing	40 × 36	56 × 36	40 × 60

Box positions (top view)



Front

A at the back; B and C in front

Place B at the front left and C at the front right of A. Their back walls touch the front wall of A.

Leave a 40 mm courtyard

The gap between the wings is the courtyard. Keep each box's joining walls. Do not overlap the boxes.

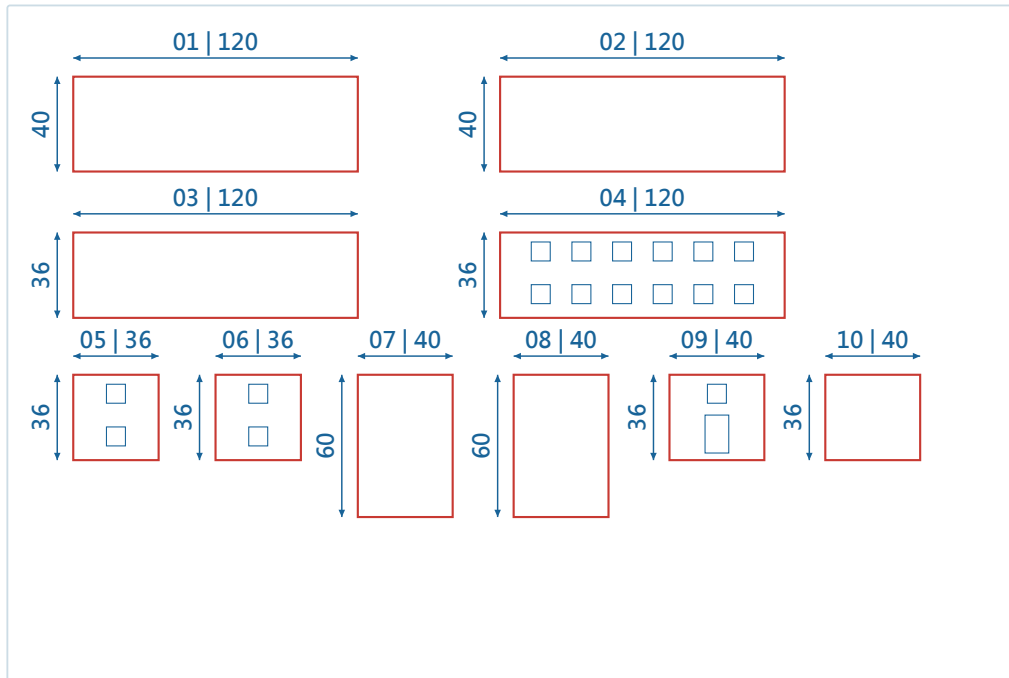
Blue lines engrave doors and windows. Leave hidden joining faces blank. Diagrams are scaled to fit; use the stated dimensions.

W5 | Template layout and part key

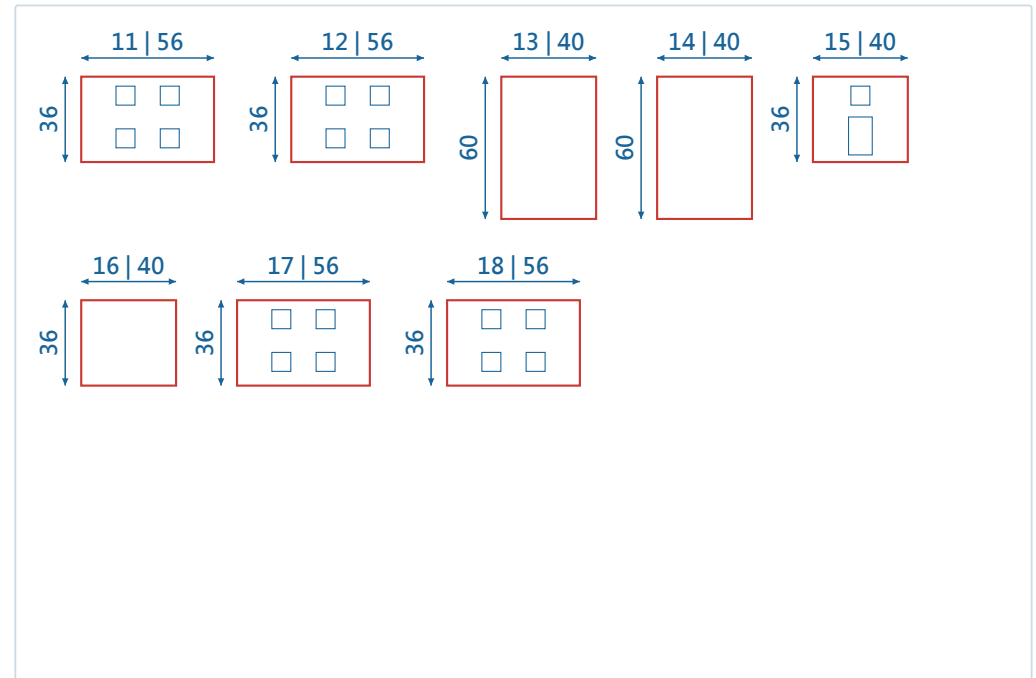
Template key

Folder: templates/W5 | 2 SVG sheet(s) | Each page: 300 × 200 mm.

Sheet 1



Sheet 2



Above: part number / width; left: height. Units: mm. Spacing adjusted for clarity; cutting SVGs have no labels.

01 A Base 120 × 40 mm

02 A Roof 120 × 40 mm

03 A Front 120 × 36 mm

04 A Back 120 × 36 mm

05 A Left 36 × 36 mm

06 A Right 36 × 36 mm

07 B Base 40 × 60 mm

08 B Roof 40 × 60 mm

09 B Front 40 × 36 mm

10 B Back 40 × 36 mm

11 B Left 56 × 36 mm

12 B Right 56 × 36 mm

13 C Base 40 × 60 mm

14 C Roof 40 × 60 mm

15 C Front 40 × 36 mm

16 C Back 40 × 36 mm

17 C Left 56 × 36 mm

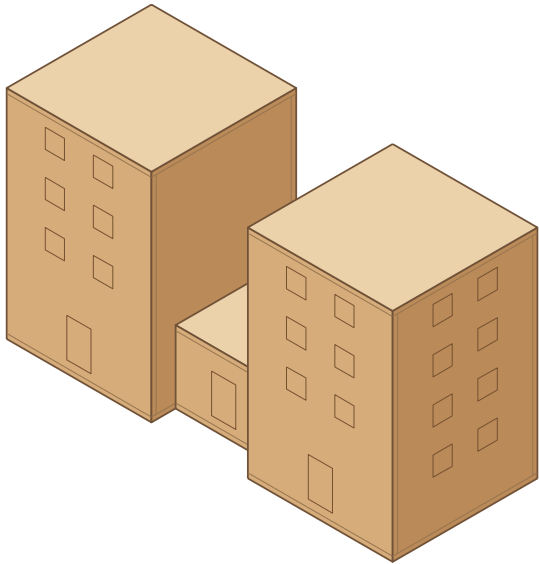
18 C Right 56 × 36 mm

W6 | Twin-block commercial building

Finished outer size: W 160 × D 60 × H 90 mm; 18 panels.



Style reference: adjust the number of windows to suit your time



Assembly diagram: use the box sizes below

Box	Use	Outer W × D × H (mm)
A	Left tower	60 × 60 × 90
B	Right tower	60 × 60 × 90
C	Entrance lobby	40 × 40 × 30

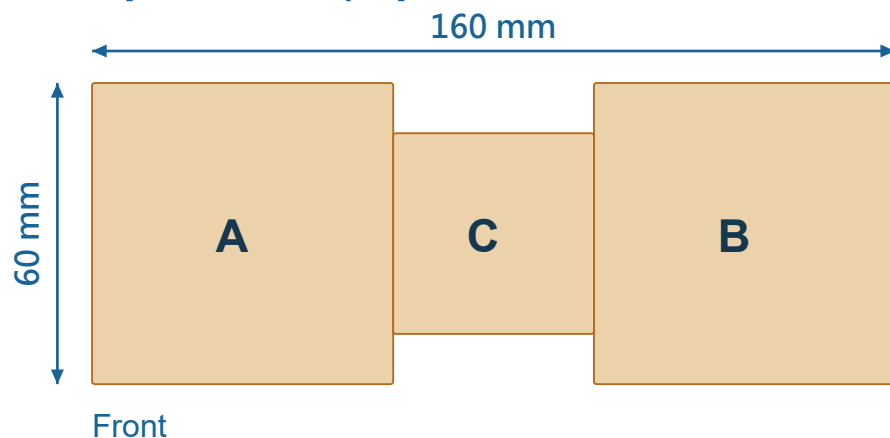
W6 | Draw the panels

Panel sizes

All sizes in mm, adjusted for 2 mm wood. Make TWO of each listed size. Do not subtract thickness again.

Box	Front / back × 2	Left / right × 2	Roof / base × 2
A Left tower	60 × 86	56 × 86	60 × 60
B Right tower	60 × 86	56 × 86	60 × 60
C Lobby	40 × 26	36 × 26	40 × 40

Box positions (top view)



A on the left; B on the right

Place lobby C in the middle. The bottoms of all three boxes sit at the same level.

Lobby inset: 10 mm front and back

Centre the 40 mm-deep lobby between the 60 mm-deep towers. Total width: $60 + 40 + 60 = 160$ mm.

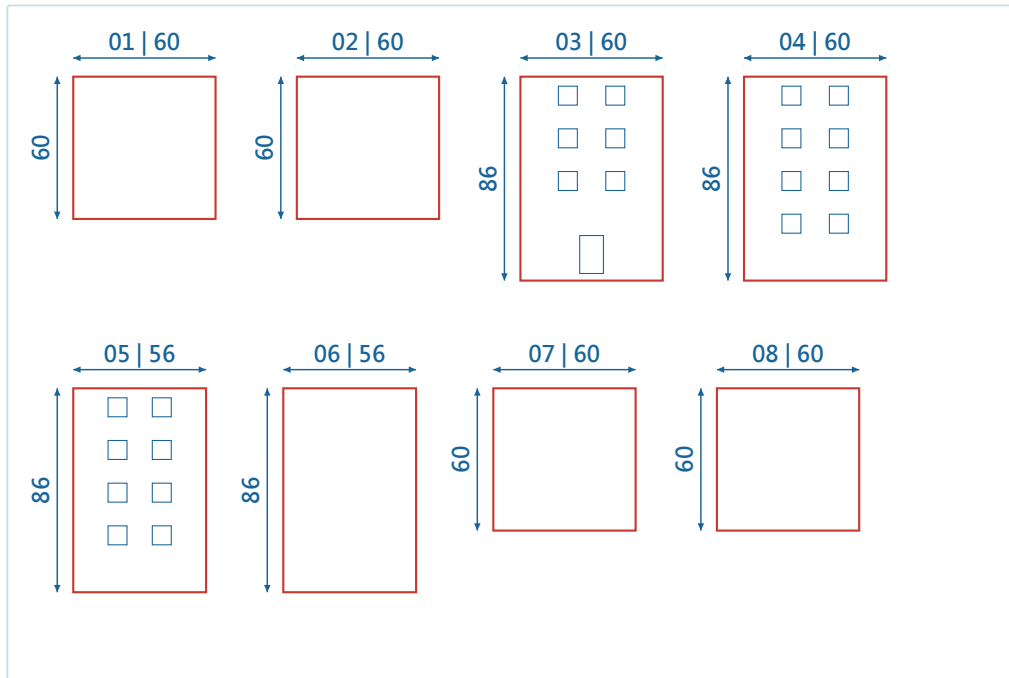
Blue lines engrave doors and windows. Leave hidden joining faces blank. Diagrams are scaled to fit; use the stated dimensions.

W6 | Template layout and part key

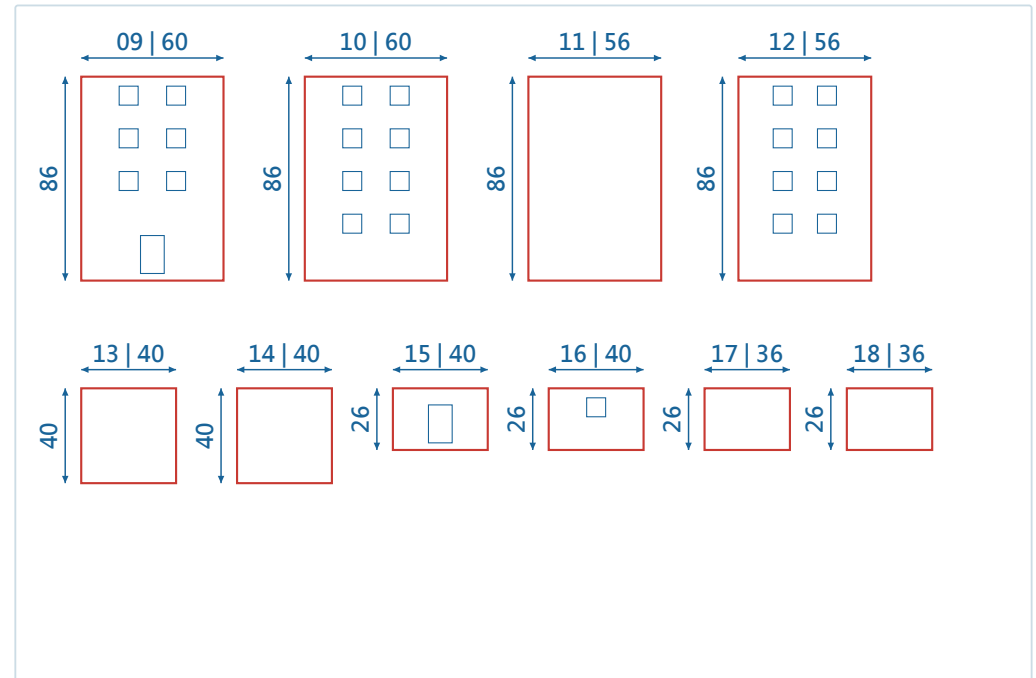
Template key

Folder: templates/W6 | 2 SVG sheet(s) | Each page: 300 × 200 mm.

Sheet 1



Sheet 2



Above: part number / width; left: height. Units: mm. Spacing adjusted for clarity; cutting SVGs have no labels.

01 A Base 60 × 60 mm

02 A Roof 60 × 60 mm

03 A Front 60 × 86 mm

04 A Back 60 × 86 mm

05 A Left 56 × 86 mm

06 A Right 56 × 86 mm

07 B Base 60 × 60 mm

08 B Roof 60 × 60 mm

09 B Front 60 × 86 mm

10 B Back 60 × 86 mm

11 B Left 56 × 86 mm

12 B Right 56 × 86 mm

13 C Base 40 × 40 mm

14 C Roof 40 × 40 mm

15 C Front 40 × 26 mm

16 C Back 40 × 26 mm

17 C Left 36 × 26 mm

18 C Right 36 × 26 mm